

Solve $y' = xy - 2y + x - 2$

$$\frac{dy}{dx} = (x-2)(y+1)$$

$$dy = (x-2)(y+1) dx$$

$$\int \frac{1}{y+1} dy = \int x-2 dx$$

$$u = y+1 \quad w = x-2$$

$$du = dy \quad dw = dx$$

$$\int \frac{1}{u} du$$

$$\int w dw$$

$$= \ln|u|$$

$$= \frac{1}{2} w^2$$

$$= \ln|y+1|$$

$$= \frac{1}{2} (x-2)^2 + C$$

$$\ln|y+1|$$

$$= \frac{1}{2} (x-2)^2 + C$$

$$e^{\ln|y+1|} = e^{\frac{1}{2} (x-2)^2 + C}$$

$$y+1 = e^{\frac{1}{2} (x-2)^2} \cdot \boxed{e^C} \quad \text{This is just a constant}$$

$$y = (e^{\frac{1}{2} (x-2)^2} \cdot C) - 1$$

$$y = e^{\frac{1}{2} (x-2)^2} - 1$$

Boundary cond: (2, 5)

$$x=2, y=5$$

$$5 = C e^{\frac{1}{2} (2-2)^2} - 1$$

$$5 = C e^0 - 1$$

$$5 = C - 1$$

$$C = 6$$

Find $f(x)$ w/ $f'(x) = f(x)$ and $f(0) = 5$

What's a function whose derivative is itself? e^x

$$f(x) = 5e^x = f'(x)$$

$$\hookrightarrow f(0) = 5e^0 = 5$$

But this isn't very rigorous

power series solutions

Suppose $f(x) = \sum_{n=0}^{\infty} C_n x^n$. Then

$$f'(x) = \sum_{n=1}^{\infty} C_n n x^{n-1}$$

$$f'' = \sum_{n=2}^{\infty} C_n n(n-1) x^{n-2}$$

Power series solutions

Suppose $f(x) = \sum_{n=0}^{\infty} C_n x^n$. Then $f'(x) = \sum_{n=1}^{\infty} C_n n x^{n-1}$

$$f'(x) - f(x) = 0 \rightarrow \left(\sum_{n=1}^{\infty} C_n n x^{n-1} \right) - \left(\sum_{n=0}^{\infty} C_n x^n \right) = 0$$

Lemma 1 : If $\sum_{n=0}^{\infty} C_n x^n = 0$, then $C_n = 0$ for all $n = 0, 1, 2, 3$

$$C_n = \frac{f^{(n)}(0)}{n!} = 0 \quad \text{But } C_{n+1} \text{ can be anything}$$

$$\left(\sum_{n=1}^{\infty} C_n n x^{n-1} \right) - \left(\sum_{n=0}^{\infty} C_n x^n \right) = 0$$

$$\hookrightarrow \left(\sum_{n=0}^{\infty} C_{n+1} (n+1) x^n \right) - \left(\sum_{n=0}^{\infty} C_n x^n \right) = 0$$

$$\hookrightarrow \sum_{n=0}^{\infty} (C_{n+1} (n+1) - C_n) x^n = 0 \quad \text{which means that } C_{n+1} (n+1) - C_n = 0 \text{ for all } x$$

So then if $f(0) = 5$ then $C_0 = 5$

$$n=0: C_1 \cdot 1 - C_0 = 0 \rightarrow C_1 = 5$$

$$n=1: C_2 \cdot 2 - C_1 = 0 \rightarrow C_2 = \frac{5}{2}$$

$$n=2: C_3 \cdot 3 - C_2 = 0 \rightarrow C_3 = \frac{5}{6}$$

$$n=1: C_2 \cdot 2 - C_1 = 0 \rightarrow C_2 = \frac{5}{2}$$

$$n=2: C_3 \cdot 3 - C_2 = 0 \rightarrow C_3 = \frac{5}{6}$$

$$\text{So } C_n = \frac{5}{n!}, \text{ then } f(x) = \sum_{n=0}^{\infty} \frac{5}{n!} x^n = 5 \sum_{n=0}^{\infty} \frac{x^n}{n!} = 5e^x$$

A Second-Order Example: Solve for $f(x)$ over the interval $(0, 1)$ with $f(0) = f(\pi) = 0$ given

If $\sum_{n=0}^{\infty} C_n x^n = 0$, then
 $C_n = 0$ for all $n = 0, 1, 2$

$$C_{n+2} = \frac{-\lambda C_n}{(n+2)(n+1)} \text{ for } n \geq 0, 1, 2, \dots$$

even $\left[C_0, C_2 = \frac{-\lambda C_0}{2 \cdot 1}, C_4 = \frac{-\lambda C_2}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{-\lambda^2 C_0}{4 \cdot 3 \cdot 2 \cdot 1} \right]$

$$C_{2n} = \frac{\lambda^n C_0 (-1)^n}{(2n)!}$$

$$f(x) = \sum_{n=0}^{\infty} C_n x^n = C_0 \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n x^{2n}}{(2n)!} + C_1 \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n x^{2n+1}}{(2n+1)!}$$

$$= C_0 \cos(\sqrt{\lambda} x) + C_1 \sin(\sqrt{\lambda} x)$$

So then $f(x) = A \sin(\sqrt{\lambda} x) + B \cos(\sqrt{\lambda} x)$

Bound cond: $f(0) = 0$ and $f(\pi) = 0$

$$f(0) = A \sin(0) + B \cos(0) = 0 \Rightarrow B = 0$$

$$f(\pi) = A \sin(\sqrt{\lambda} \pi) + 0 = 0$$

W/ this bound cond, $f(x)$ is just a sin function?

If $A=0$, $f(x)=0$. If $\sin(\sqrt{\lambda} \pi) = 0$, $\sqrt{\lambda} \pi = n\pi$ where $n = 0, 1, 2, \dots$ or $\lambda = n^2 = 0, 1, 4, 9, 16, \dots$

A different bound cond: $f'(0) = 0$ and $f'(\pi) = 0$

$$f'(0) = A \sqrt{\lambda} \cos(0) - B \sqrt{\lambda} \sin(0) = 0 \Rightarrow A = 0$$

pulled out via chain rule?

$$\text{So } A = 0$$

$$f'(\pi) = 0 - B \sqrt{\lambda} \sin(\sqrt{\lambda} \pi) = 0$$

W/ this bound cond, $f(x)$ is just a cos?

If $B=0$, $f(x)=0$

If $\sin(\sqrt{\lambda} \pi) = 0$, $\sqrt{\lambda} \pi = n\pi$
 $n = 0, 1, 2, \dots$
 or $\lambda = n^2 = 0, 1, 4, 9, 16, \dots$

$f''(x) = -\lambda f(x)$. $\lambda = \text{any real number}$

$$\hookrightarrow \sum_{n=2}^{\infty} C_n (n)(n-1) x^{n-2} + \left(\sum_{n=0}^{\infty} C_n \lambda x^n \right) = 0$$

$$\hookrightarrow \sum_{n=0}^{\infty} C_{n+2} (n+2)(n+1) x^n + \left(\sum_{n=0}^{\infty} C_n \lambda x^n \right) = 0$$

$$\hookrightarrow \sum_{n=0}^{\infty} (C_{n+2} (n+2)(n+1) + \lambda C_n) x^n = 0$$

odd $\left[C_1, C_3 = \frac{-\lambda C_1}{3 \cdot 2 \cdot 1}, C_5 = \frac{-\lambda C_3}{5 \cdot 4} = \frac{-\lambda^2 C_1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \right]$

Power Series Solutions: For more difficult differential equations (higher order, not separable), we can employ power series to represent our solution.

Example: Find the first four terms of the power series of $f(x)$ solving

→ Second order (two derivatives) → need 2 boundary conditions
 $f''(x) = xf(x)$. And $f(0) = 1$, $f'(0) = 1$

Let $f(x) = \sum_{n=0}^{\infty} C_n x^n = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots$
 Given bound cond...
 $C_0 = 1$
 $C_2 = 1$
 $C_2 = 0$
 $C_3 = \frac{1}{6}$
 $C_4 = \frac{1}{22}$
 $C_5 = 0$

$f'(x) = \sum_{n=2}^{\infty} C_n n x^{n-1} = C_2 + 2C_3 x + 3C_4 x^2 + \dots$
 for $f'(0) = 1$, C_2 must be 1
 for $f'(0) = 1$, C_2 must be 1

these can be gotten simply, but
 for the rest you need diff eqs

$f''(x) = \sum_{n=2}^{\infty} C_n n(n-1) x^{n-2} = 2C_2 + 6C_3 x + \dots$

Because $f''(x) = xf(x)$
 $2C_2 + 6C_3 x + \dots = x(C_0 + C_1 x + C_2 x^2 + \dots)$
 $= C_0 x + C_1 x^2 + C_2 x^3 + \dots$
 No constant term here

So $2C_2$ must be 0 so $C_2 = 0$
 $2C_2 + 6C_3 x + \dots = 0 + C_0 x + C_1 x^2 + C_2 x^3 + \dots$
 $2C_2 = 0$ $6C_3 = C_0 \Rightarrow$ so $C_3 = \frac{1}{6} C_0 = \frac{1}{6}$

$2C_2 + 6C_3 x + 22C_4 x^2 = 0 + C_0 x + C_1 x^2 + C_2 x^3 + \dots$
 $+ 20C_5 x^2 + \dots$
 $22C_4 = C_1 \Rightarrow C_4 = \frac{1}{22} C_1$
 $20C_5 = C_2 \Rightarrow C_2 = 0$ so $C_5 = 0$

$f(x) = 1 + x + 0x^2 + \frac{1}{6}x^3 + \frac{1}{22}x^4 + 0x^5 + \dots$

Example: Find the first three terms of the power series of $f(x)$ solving

$$f'''(x) = 2f'(x) + xf(x).$$

Example: Solve for $f(x)$ with $f(0) = 1$ and $f'(0) = 1$ given

$$(x^2 + 1)f''(x) = 4xf'(x) - 6f(x).$$

Find first 5 terms w/o bound conds

$$\text{Let } f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

$$\text{Let } f'(x) = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$\text{Let } f''(x) = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

$(x^2 + 1)f''(x) = x^2 f''(x) + f''(x)$ How did we get this - Just by distributing?

$$= (2c_2 x^2 + 6c_3 x^3 + 12c_4 x^4 + \dots) + (2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots)$$

$$4xf'(x) - 6f(x) =$$

$$(4c_1 x + 8c_2 x^2 + 12c_3 x^3 + \dots) - (6c_0 - 6c_1 x - 6c_2 x^2 - 6c_3 x^3 - \dots)$$

$$2c_2 = -6c_0 \quad \text{const term}$$

$$6c_3 = 4c_1 - 6c_2 \quad \text{linear term}$$

$$6c_4 = -2c_1$$

$$2c_2 x^2 + 12c_4 x^2 =$$

$$8c_2 x^2 - 6c_2 x^2$$

$$\hookrightarrow 2c_2 + 12c_4 = 2c_2$$

$$\hookrightarrow c_2 + 12c_4 = c_2$$

$$\text{So } c_4 \text{ must be 0}$$

Quadratic term

$$\begin{aligned} 6c_3 + 20c_5 &= 6c_3 & \text{cubic term} \\ c_3 + 20c_5 &= c_3 \\ \text{So } c_5 &= 0 \end{aligned}$$

$$c_0 =$$

$$c_1 =$$

$$c_2 =$$

$$c_3 =$$

$$c_4 = 0$$

$$c_5 = 0$$

everything else will be 0

Now with the boundary conditions we can solve for c_0, c_1, c_2 , and c_3

$$f(0) = 1$$

$$\hookrightarrow 1 = c_0 + \cancel{c_1 x} + \cancel{c_2 x^2} + \dots$$

$$c_0 = 1$$

$$f'(0) = 1$$

$$1 = c_1 + \cancel{2c_2 x} + \cancel{3c_3 x^2} + \dots$$

$$c_1 = 1$$

$$2c_2 = -6c_0 \quad \text{const term}$$

$$6c_3 = 4c_1 - 6c_2 \quad \text{linear term}$$

$$6c_4 = -2c_1$$

$$2c_2 = -6$$

$$c_2 = -3$$

$$6c_3 = -2$$

$$c_3 = -\frac{1}{3}$$

$$c_0 = 1$$

$$c_1 = 1$$

$$c_2 = -3$$

$$c_3 = -\frac{1}{3}$$

$$c_4 = 0$$

$$c_5 = 0$$

everything else will be 0

Since all the remaining terms are 0 we can get a full function

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = 1 + 1x - 3x^2 - \frac{1}{3}x^3$$